



IT 474 - Network Security
Lab#1

Lab Objectives:

1. To establish the context of the lab activities and create groups.
2. To introduce and install a VirtualBox
3. To introduce and install Ubuntu Linux.
4. Establish a virtual LAN

Lab Material used:

1. Web Browser with Internet Access.
2. VirtualBox software.
3. The provided *NetworkSecurity* OVA file (around 4 GB)

Part1: Lab Content and Plan [10 Mins.]

Slides Discussion

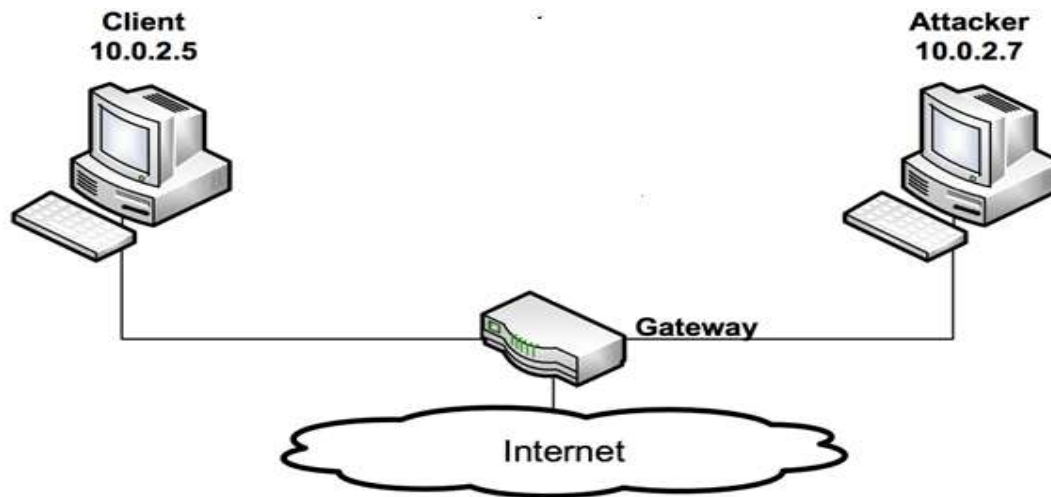
Part2: Installation [30 Mins.]

- Download and install the [VirtualBox](#) from the internet
- Download and install the provided Ubuntu OVA file by doing the following:
 1. Open **VirtualBox** software
 2. Download Ubuntu OVA file from:
https://drive.google.com/file/d/1LHGG4QXtLkCGPZzllB83D53zSm97DiIZ/view?usp=drive_open
 3. Import the OVA file as an appliance: **VirtualBox** -> **File** -> **Import Appliance** -> navigate to the downloaded OVA file -> **Import**. You may want to refer to [3].
 4. After the importing process completed, select the newly imported VM named *NetworkSecurity* from the left pane of VirtualBox Window.
 5. Change the USB to 1.1:
click on your VM -> Settings -> Ports -> USB -> USB 1.1 -> OK
 6. Rename your VM to Your LMS registered Team Code
click on your VM -> Settings -> General -> Basic -> change the name from *Network Security* to your team code.
 7. Start the VM you have just renamed.

Part3: Network Setup : [15 min]

Each VM will take about **10 GB** of your hard disk space. Each VM will need at least **1024 MB** of RAM. 1 CPU is enough. You need to end up with two machines as will be described below.

Set up 2 virtual machines on the same host computer on the same LAN as described in the figure below. One computer is used as **Attacker (VM has been imported in part2)**, the second computer is used as the **Victim**. Please note the figure is conceptual only. I.e. you don't have to have the exact same IP addresses.



Here is a suggestion to the required steps to accomplish the task and some helpful resources. Please Feel free to do and use whatever you want as long as you will be able accomplish the task.

Suggested steps:

1. Study your needs.
2. Find the best machine on your team to suit those needs.
3. Decide whether to make a full clone or a linked clone.
4. Set up a virtual LAN [1]. You want to have a NAT NETWORK.
5. Connect your VM to that LAN [1].
6. Clone your VM [2].

Evaluation:

Screenshot#1: The two VMs named as required.

Screenshot#2 and 3: The assigned IP address for each of VMs.

Suggested Resources:

- [1] [How to create multiple NAT Networks in VirtualBox | TechRepublic](#)
- [2] <https://dirkstrauss.com/clone-virtualbox-vm/>
- [3] OVA Installation Tutorial <https://itsfoss.com/install-kali-linux-virtualbox/>

Youtube videos:

- <https://www.youtube.com/watch?v=Hva8lsV2nTk>
- <https://www.youtube.com/watch?v=DzmUOeFdc-w>